

Project Overview

Project Goals:

- Create an autonomous racing agent that can consistently hot lap within top 95% of human times (thats better then me >:C)

Project Justification:

- Model can be used to test how changes in vehicle setup effects lap times
- Model can be used as reference for humans on where time can be gained throughout a lap

Week 1 Goals:

- Create a system for gathering telemetry data from Assetto Corsa
- Collect data at Indianapolis Motor Speedway (<https://assetto corsa.club/mods/tracks/indianapolis-motor-speedway.html>) driving BMW Z4 GT3 (Assetto Corsa Base Game)
- Train reinforcement model that can successfully complete a lap

Terminology:

- FR: front right, FL: front left, RR: rear right, RL: rear left

Inputs:

- Position [x, y, z]
- Velocity [x, y, z]
- Acceleration [x, y, z]
- Tire Temperatures [FR, FL, RR, RL]

Outputs:

- Steering Angle -360 <-> 360
- Brake 0 <-> 1
- Throttle 0 <-> 1

Meeting Reports

1/16/2026 Meeting:

What we did:

- Researched other projects similar to ours
- Decided first imitation model will be on Indianapolis using BMW Z4 GT3
- Used ACTI to collect telemetry data
 - Leo is working on proprietary data collection method to streamline the data collection process

What to do next:

Leo:

- Implement data collection and exporting into .csv

Lawrence:

- Research and work on creating imitation model given .csv

Week of 1/19:

- Created telemetry collection application
- Recorded 15 laps of Indianapolis using Z4
- Researched lowest latency input method
 - Settled on simulating an xbox controller using vigenbus

1/23/2026 Coleman Meeting:

- Last week, goal was to get data from Assetto Corsa simulator for a circular track.
- Lawrence got this data by driving around a circular track for a few laps. Data is in CSV (30Hz) format with a lot of information (steering, tires, actuators, etc.)
- Using something to simulate an Xbox controller because there doesn't appear to be an easy way to just inject the steering data directly in the simulator.
- Leo started looking at potential models to use.

TODO:

- Test virtual xbox controller on game, try to use the telemetry data to control the virtual xbox controller and replay the car moving around the track (compare with new telemetry)

Weeks 4-5:

- Recreated ACGym results
 - Was able to replicate their results
- Learned how to run "headless" mode
 - Lower resolution and minimize application

TODO:

- figure out how to record data
- take distance to racing line out of model's reward function
- test headless more and see if we can speed up game or run multiple instances
- automatic gearbox loses A LOT of time

- Maybe add gearbox controls to model output

Week 7 TODO:

get our own SAC model working (put it in our own environment)

- add shifting to model outputs

get server access and train on server

3/20:

- Got it working in our own environment but trained for 8 hrs and model just stays in place and brakes
 - Unsure why
- have a lot of trouble training it on Linux server
 - Vigembus doesn't work on Linux
 - Shared memory is different on Linux
 - Don't know if there is value from training on server if we can't speed up training anyways due to assetto corsa physics clock
 - Could get value from parallelization but that is a lot of implementation
- What to do for next time
 - Get baseline model working
 - Try removing curriculum
 - Note: unsure if the reason model stays in place braking and steering is curriculum but it is most likely culprit
 - Once that works, try getting curriculum working again
 - Look into parallelization and possibly migrating / forking the repo to work with the server's linux system
 - Could also keep current implementation and just run windows vm on linux server

3/27:

What we did:

- made distilled version of assetto corsa gym
 - One using vjoy and one using vigembus
- The problem was that we have to do 1 inference per physics packet (before we were just spamming a bunch of inputs per physics packet since physics only updates 30hz)
- Also noticed the abs and traction control being on made the model's gas and brake outputs a lot cleaner (holds only gas fully on straights and fully brakes at start of braking zones vs messy brake and throttle output when abs and tc are off)

We were able to train a working model that performs the same, if not better than assetto corsa gym. The key problem that we were missing this whole time was to sync each model inference with each physics packet (literally wait for it to be received, and only then we do model inference). Before, the model was sending outputs at 30hz (which is the same hz that the physics update runs at) but they weren't exactly synced.

Also when we trained this current working model, we forgot to turn traction control and ABS off in the car. This actually seems to have helped the model because its braking and throttle inputs were a lot cleaner. It would fully hold the gas on straights and not tap the brakes (which was a problem with assetto corsa gym). I was thinking we could add this to our curriculum, where the model starts training with TC and ABS but it gets less and less and eventually turned off.

We're still in the process of getting the model working on the server but we should be able to. There are people online who have gotten assetto corsa + content manager working on linux. We had some problems getting firefox working but we sorted that out (it wasn't opening on the correct display).

What to do this week:

- Add back curriculum learning
 - slowly lower reward on racing line adherence
 - slowly decrease TC and ABS
- Get model training on Linux server
 - Make Linux fork of the project
- Give training examples

4/10 Meeting:

What we did this week:

- Trained first 3 models in "Model Reports" tab
- Learned that ABS and TC help model performance (training with no ABS hindered the model from learning how to brake)
- Model was also able to successfully learn shifting (not completely optimal yet but was on the way if had more training time)

What to do this week:

- Add TC and ABS to curriculum (start at tc and abs 2 or 3, then slowly lower until off)
- Record and give demonstrations to model

4/17 Meeting:

- Try best at getting model with shifting working
- Aim for next week to be able to train final good model
- Look into different demonstration pretraining methods

4/24 Meeting:

- This week we should both just train our models for the whole week
- Draft github page and send to Dr. Coleman
 - Make sure to include all the things we tried that didnt work, along with the things that did

- Don't just post results, post the whole process of creating for future people who either continue this work or if we continue this work

Related Links

Project GitHub: <https://github.com/Law47/AutonomousRacingAgent>

NVIDIA Presentation on Racing AI: <https://www.youtube.com/watch?v=2oYJQqepRSM>

Similar, but failed, student project: <https://www.youtube.com/watch?v=hJRGfN3b2EE>
<https://github.com/Jurredr/ACRL>

Competitor: <https://assetto-corsa-gym.github.io/>
https://github.com/dasGringuen/assetto_corsa_gym

Reinforcement Learning Algorithm (SAC): <https://arxiv.org/abs/1801.01290>

Model Reports

ACGym: 1:52:61 SAC and 1:51:64 SAC fD

Note: TC and ABS Factory defaults are TC 3 and ABS 5

4/9 Report:

Trained overnight ~ 11 hours, 278 laps

Mode: Same as acgym except with curriculum that slowly lessens racing line reward

TC and ABS Factory defaults

Result: mid 1:56 average, lowest 1:53:976 (Note: these lap times were all invalid because model wasn't braking enough for first corner)

4/10 Report TRAINING WITH SHIFTING:

Trained overnight ~8 hours

Mode: TC and ABS off, Auto-shifting Off (Shifting added to model outputs)

Added penalty for shifting into neutral and reverse gear

Result: Model learnt how to shift but seems worse at taking some corners. The model may just need more training since it seems so sometimes shifts at the right spots. For example sometimes it shifts down properly while sometimes it does not shift down and misses out on potential seconds. Also the model shifts up sometimes suboptimally. The model also takes the corners worse than the model that learnt without shifting.

4/10 Report:

Trained overnight ~9 hours

Mode: Can only brake or gas one at a time

TC and ABS off

Reset penalty added to curriculum (after 9 hours it's only being punished a little so maybe increase the scaling along; increase racing line reward decrease curriculum too)

Result: Was going around the track but no recorded laps due to going off track on any heavy braking corners. Doesn't seem to brake at all for some corners. When it brakes it locks up front tyres due to no ABS which might be why it's scared to brake.

4/10 Report:

Trained during the day ~9 hours (533 minutes)

Mode: Manual Shifting, TC: 3/10, ABS: 5/10

Result: The model stands still and gear shifts randomly.



<https://imgur.com/a/DRn2jZj>

4/11 Report:

Trained for ~24 hours, ~1.35 million steps.p

Mode: Manual Shifting, TC: 3/10. ABS: 5/10

Result: Model seems to drive decently on some corners, while on other corners, it hardsends and overshoots, causing a reset. The shifting behaviour seems inconsistent like for example, during some sections of the track it properly shifts but in other cases it stays on neutral or 1st gear. It even goes into reverse gear sometimes, which should never happen.

4/15 Report:

Trained for ~8 hours

Mode: Demonstration Learning First, Manual Shifting, TC 3/10, ABS 5/10

Result: Model seemed to be in correctly learning how to shift. After investigation there were flaws in how the demonstration data denoted a shift up and shift down.

4/16 Report:

Trained for ~3 Hours

Mode: 10 epochs of Demonstration Data Behavior Cloning, Manual Shifting, TC 3/10, ABS 5/10

Result: Cut off model as it seemed too incapable of driving

4/16 Report:

4/17 Report:

Trained for ~8 hours

Mode: Demonstrations (vanilla SAC updates on demo buffer (weak, not good)), Manual Shifting, TC and ABS off, Can only brake or gas one at a time

Result: Made it's way around the track but had to be reset on any hard braking corners. Model seemed to shift decently and was being downshift protected when entering a corner (meaning it knows it has to be in a lower gear but didn't learn it has to brake to be able to downshift to that lower gear)

4/17 Report

Trained for ~8 hours

Mode: Demonstrations (behaviour cloning), Manual Shifting, TC: 3/10, ABS: 5/10

Result: Model drives erratically steering all over the place. The model also seems to shift in consistently.

4/20 Report

Trained for ~8 Hours

Mode: Demonstrations (2 epochs (tried 5 earlier with same results)), manual shifting, no TC no ABS

Result: Model just full gas and turns right while shifting between 1-3 gears

4/20 Report

Trained for ~24 hours, 2.8 million steps

Mode: Demonstrations (5 epochs), With Offline vs Online Buffer split setup, manual shifting, TC + ABS on.

Result: Model performs a lot better with shifting, as expected, given it has been given more steps. However, the model seems to have found that the optimal way to get the most rewards is to dive bomb two of the corners of the track to get the most reward out of maintaining max speed, ignoring the going out of track penalty. This implies that my reward function needs to be adjusted accordingly

4/22 Report

Trained for ~8 hours, 750000 steps but in AC timer shows 100 minutes?

Mode: Demonstrations (2 epochs), with 50% offline and 50% online, manual shifting, TC and ABS off

Result: Holding the brake and turning the wheel

Note: There may be something wrong with my setup as the AC timer doesn't align with expected time.

4/26 Report

Trained for ~24 hours, 2million+ steps

Mode: Demonstrations off/ AutoShifting for 1.5 million steps, Manual shifting for the rest.

Separated Shifter and Driver Models.

Result: Model performed really well when it had auto-shifting enabled. It was able to complete a valid lap with the time of 1:55:35 which is our best valid lap so far. However when the model attempted to learn manual shifting it quickly fell apart and was unable to complete a single lap.

4/28 Report

Trained for ~11 hours, 1.4 million steps

Mode: Demonstrations on, Offline Sampling: 15% to 0.02% over 4 million steps, Demonstration pretrain_epochs: 1, Auto Shifting -> Manual Shifting Curriculum.

Result: Model seemed to really like the brake pedal to the point where the car would not drive. The result was not satisfactory.